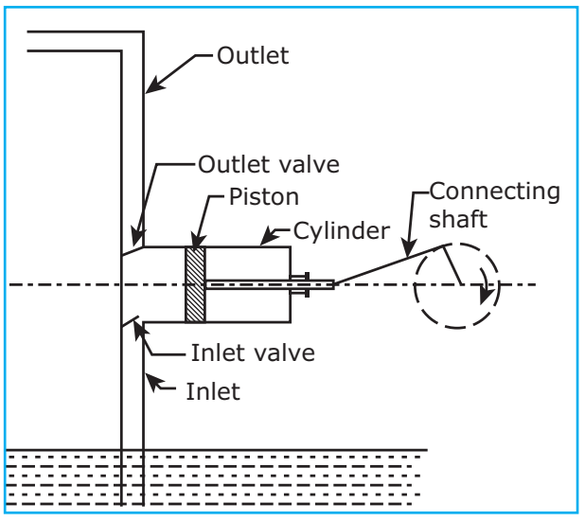
8.9 Reciprocating pump

Reciprocating pump is one in which the liquid is pumped by the reciprocating action of the moving part known as a piston or a plunger inside a cylinder.

Components of a reciprocating pump

1. Cylinder
2. Piston
3. Connecting rod
4. Crank
5. Suction pipe
6. Suction valve
7. Delivery pipe
8. Delivery valve



Reciprocating pump

Cylinder

Cylinder is a hollow part inside of which is accurately machined. Piston reciprocates within the cylinder.

Piston

Piston is the reciprocating part of the pump. A connecting rod is connected to it on one side.

Connecting rod

Connecting rod is attached to piston on one end and with a crank on the other end. It converts the rotary motion of the crank into reciprocating motion of the piston.

Working principle of a reciprocating pump

When the crank is made to rotate by means of prime mover, the piston reciprocates within the cylinder to pump out the liquid.

Suction stroke

As shown in fig, when the piston starts moving towards right, a vacuum is created on the left side of the cylinder. The suction valve opens and the liquid is sucked through the suction pipe into the cylinder. The delivery valve remains closed at this stage.

Deliver stroke

Due to the rotation of the crank, piston is moved towards left side. The piston exerts pressure on the liquid inside the cylinder and the delivery valve opens. The liquid is thrown through the delivery pipe with pressure. At this stage, the suction valve remains closed.

On continuous reciprocation of the piston, liquid is sucked and pumped continuously.

Application of reciprocating pump

- Pumping sea water, ship and marine field
- Air pumping devices
- Filling water in small grade boilers
- Oil pumping

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